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NUMERICAL MODELING OF ENERGY HARVESTING WITH PIEZOELECTRIC MATERIALS USING NON-SMOOTH SYSTEMS

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Abstract: *The world increasingly relies on autonomous devices that require energy to function. At the same time, the energy transition towards clean energy solutions is also gaining increasing relevance. In this sense, vibration-based energy harvesting is an interesting approach, but it often faces the limitation of not operating efficiently under variable excitation frequencies, such as those typically found in mechanical and structural systems. One way to expand the operating range of these devices is to introduce nonlinearities into the system by incorporating magnets or impact-generating elements. This work investigates the dynamic behavior and energy harvesting performance of a piezoelectric cantilever beam exhibiting Duffing-type and nonsmooth nonlinearities. The system considered consists of an aluminum cantilever beam with a surface-bonded piezoelectric patch, excited harmonically and constrained to vibrate between mechanical end-stops and including magnetic forces using magnets that can be simulated as cubic stiffness contribution. An electromechanical model is formulated by combining the linear piezoelectric constitutive relations with a single-degree-of-freedom mechanical model including a Duffing term and two mechanical end-stops. The resulting set of coupled nonlinear ordinary differential equations, describing the displacement and the voltage across a resistive load, is implemented numerically with Fourth order Runge-Kutta method. A systematic parametric study is carried out to evaluate the influence of mechanical parameters associated with the nonlinear force provided by the magnet and end-stops impacts, first separately and after combining the effects, in order to verify the best performance. The numerical results indicate that the introduction of Duffing-type nonlinearity with the traditional non-smooth systems can significantly broaden the effective frequency bandwidth of operation and increase the average harvested power over a range of excitation conditions. These findings contribute to the design of robust, broadband piezoelectric energy harvesting devices based solely on computational modeling and provide guidance for future experimental validation.*

Palavras-chave: Energy harvesting, Piezoelectric materials, nonlinear dynamic, Duffing type, Non-smooth systems

1. INTRODUCTION

The increasing demand for low-power autonomous devices, such as wireless sensors and embedded monitoring systems, has intensified research on vibration-based energy harvesting technologies. Among the available mechanisms, piezoelectric transducers are particularly attractive due to their direct electromechanical coupling, structural simplicity, and ease of integration into mechanical systems. Comprehensive reviews have highlighted the versatility of piezoelectric energy harvesting across civil, aerospace, and microsystems applications (Liu et al., 2018; Sezer & Koç, 2021; Safaei et al., 2019). The fundamental distributed-parameter and reduced-order electromechanical modeling frameworks for cantilever-based harvesters were systematically established by Erturk and Inman (2011) and remain a cornerstone in the field.

In practical implementations, piezoelectric harvesters are typically tuned to operate near their fundamental resonance frequency in order to maximize strain-induced voltage. However, linear resonant systems inherently exhibit narrow operational bandwidth, which significantly limits their effectiveness under broadband, low-frequency, or time-varying

environmental excitations (Harne & Wang, 2013). Recent design studies continue to emphasize the need for passive bandwidth broadening strategies capable of maintaining performance away from the primary linear resonance (Xu et al., 2022; Zhou et al., 2022).

One promising approach to overcome this limitation is the introduction of structural nonlinearities. Within the classical framework of nonlinear dynamics and chaos, the Duffing oscillator serves as a reference model for systems exhibiting cubic stiffness nonlinearities, multistability, hysteresis, and jump phenomena (Savi, 2014; Savi, 2017). Such nonlinear characteristics may substantially modify the frequency response and enhance energy conversion efficiency. Erturk and Inman (2011) demonstrated that bistable Duffing-type harvesters operating on high-energy interwell orbits can significantly increase electrical power output. Expanding upon this concept, multistable configurations have been proposed as an effective strategy for further bandwidth enhancement. Costa, Monteiro and Savi (2024) showed that properly tuned multistable nonlinear harvesters can improve average power generation in compact devices. Additional experimental and numerical investigations have confirmed the relevance of multi-stability for performance enhancement (Mitura et al., 2024; Caetano et al., 2025), while advanced architectures such as quasi-zero stiffness and multi-well systems have been introduced to reduce activation thresholds and improve low-frequency harvesting (Hua et al., 2025; Kloda, 2026).

In parallel with smooth cubic nonlinearities, non-smooth dynamics arising from unilateral contact and vibro-impact mechanisms have attracted increasing attention. Unlike continuous Duffing-type nonlinearities, impact-based systems introduce piecewise-smooth behavior and abrupt stiffness transitions, potentially activating additional dynamic regimes and modifying global response characteristics. Experimental investigations have shown that non-smooth cantilever harvesters may exhibit bandwidth enlargement and complex dynamic behavior under periodic excitation (Li et al., 2026). Similar improvements have been reported in stopper-assisted and Hertzian contact-based configurations (Shahsavari & Ashory, 2024; Li et al., 2024), as well as in impact-driven frequency up-conversion strategies (Aceti et al., 2023; Peng et al., 2024). Theoretical studies addressing unilateral barriers and nonsmooth energy harvesting mechanisms further indicate that contact nonlinearities can substantially alter displacement amplitude and harvested power under both deterministic and stochastic excitations (Su et al., 2020; Ai et al., 2019).

Despite the extensive body of literature on smooth Duffing-type nonlinearities and non-smooth impact mechanisms considered independently, comparatively fewer studies examine their combined interaction within a unified electromechanical modeling framework. From the perspective of nonlinear dynamics, the coexistence of cubic stiffness effects and intermittent contact may produce complex transitions between coexisting attractors, modify global bifurcation structures, and influence the effective harvesting bandwidth (Savi, 2017). The implications of this hybrid nonlinear behavior for energy harvesting optimization remain insufficiently characterized.

In this context, the present work investigates the dynamic response and energy harvesting performance of a piezoelectric cantilever beam modeled as a single-degree-of-freedom electromechanically coupled system incorporating both Duffing-type cubic stiffness and unilateral mechanical end-stops. The formulation combines classical piezoelectric constitutive relations (Erturk & Inman, 2011) with a non-smooth dynamic representation capable of capturing impact interactions. Through systematic parametric analyses, the isolated and combined effects of smooth and non-smooth nonlinearities are evaluated in terms of vibrational response and average harvested power. The results contribute to the understanding of hybrid nonlinear strategies for broadband piezoelectric energy harvesting and provide insights for the design of robust and computationally optimized nonlinear devices.

2. MATHEMATICAL MODEL

In this work, a mathematical description of a non-smooth energy harvesting system incorporating the effects of magnetic forces is presented. The physical system under investigation consists of an aluminum cantilever beam with a piezoelectric layer, where one end is clamped while the other remains free, see Figure 1. The clamped end is connected to an electromechanical shaker capable of providing different types of mechanical excitation.

Simultaneously, two mechanical stoppers are placed near the free end of the beam, introducing nonlinearity into the system through impact interactions. These stoppers are mounted on a rigid support structure, allowing the gaps between the beam and the stoppers, denoted by g_1 and g_2 , to be varied. In addition to the mechanical stoppers, magnetic elements are introduced into the system. One magnet is attached to the free end of the piezoelectric beam, while another is positioned in its vicinity. The magnets are arranged in a repulsive configuration, generating magnetic forces that render the system bistable and give rise to complex dynamic behavior. In this configuration, the horizontal distance d between the fixed magnet and the magnet attached to the piezoelectric beam can be adjusted.

- Contact with superior stopper 1:

$$m\ddot{z} + (c + c_{s1})\dot{z} + k_p(z)z + k_{s1}(z - g_1) - \theta V = -m\ddot{u} \quad \text{if } z \geq g_1 \quad (2)$$

- Contact with inferior stopper 2:

$$m\ddot{z} + (c + c_{s2})\dot{z} + k_p(z)z + k_{s2}(z - g_2) - \theta V = -m\ddot{u} \quad \text{if } z \geq g_2 \quad (3)$$

where $k_p(z) = k_{p1} + k_{p2}z^2$. The mechanical nonlinearity is represented by a Duffing-type restitution force that describes the general behavior expressed by $f(z) = k_{p1}z + k_{p2}z^3$ and the potential energy $U(z) = \frac{1}{2}k_{p1}z^2 + \frac{1}{4}k_{p2}z^4$.

- The electrical differential equation is given by:

$$\theta\dot{z} + C_p\dot{V} + \frac{1}{R_l}V = 0 \quad (4)$$

The instantaneous electrical power is defined as $P = V^2/R_l$, and the average output (P_{out}) power is defined as

$$P_{out} = \sqrt{\frac{1}{t} \int_0^t (V^2/R_l)^2 dt} \quad (5)$$

A dimensionless analysis of piezoelectric vibration energy harvester mathematical model is now in focus. Hence, consider spatial and electrical new coordinates as $x = z/l$, $\gamma_i = g_i/l$ and $v = V/\hat{V}$ where l is a reference length and $\hat{V}$ is a reference voltage. A reference frequency is defined as $\omega_0 = \sqrt{|k_{p1}|/m}$ and $\tau = \omega_0 t$ is the nondimensional time. Using $\zeta = \frac{c}{m\omega_0}$, $\zeta_{s1} = \frac{c_{s1}}{m\omega_0}$, $\zeta_{s2} = \frac{c_{s2}}{m\omega_0}$, $\alpha = \frac{k_{p1}}{m\omega_0^2}$, $\beta = \frac{k_{p2}l^2}{m\omega_0^2}$, $\epsilon = \frac{\theta\hat{V}}{ml\omega_0^2}$, $\chi = \frac{A}{l\omega_0^2}$, $v_{s1} = \frac{k_{s1}}{m\omega_0^2}$, $v_{s2} = \frac{k_{s2}}{m\omega_0^2}$, $\sigma = \frac{\theta l}{C_p\hat{V}}$ and $\rho = C_p\omega_0 R_l$, the motion and electrical differential equations can be rewritten as follows:

- No contact (dimensionless form):

$$\ddot{x} + \zeta\dot{x} + \alpha x + \beta x^3 - \epsilon v = -\chi \sin(\bar{\omega}\tau) \quad \text{if } x < \gamma_1 \text{ and } x > \gamma_2 \quad (6)$$

- Contact with superior stopper 1 (dimensionless form):

$$\ddot{x} + (\zeta + \zeta_{s1})\dot{x} + \alpha x + \beta x^3 + v_{s1}(x - \gamma_1) - \epsilon v = -\chi \sin(\bar{\omega}\tau) \quad \text{if } x \geq \gamma_1 \quad (7)$$

- Contact with inferior stopper 2 (dimensionless form):

$$\ddot{x} + (\zeta + \zeta_{s2})\dot{x} + \alpha x + \beta x^3 + v_{s2}(x - \gamma_2) - \epsilon v = -\chi \sin(\bar{\omega}\tau) \quad \text{if } x \geq \gamma_2 \quad (8)$$

- Electrical differential equation (dimensionless form):

$$\sigma\dot{x} + \dot{v} + \frac{v}{\rho} = 0 \quad (9)$$

where the dimensionless non-linear stiffness is $\hat{k}_p(x) = \alpha + \beta x^2$. The dimensionless restitution force that describes the general behavior expressed by $\hat{f}(x) = \alpha x + \beta x^3$ and the potential energy $\hat{U}(x) = \frac{1}{2}\alpha x^2 + \frac{1}{4}\beta x^4$. The instantaneous dimensionless electrical power is defined as $\hat{P} = v^2/\rho$, and the dimensionless average output power ($\hat{P}_{out}$) and the dimensionless average displacement ($\hat{x}$) defined as

$$\hat{P}_{out} = \sqrt{\frac{1}{\tau} \int_0^\tau (v^2/\rho)^2 d\tau} \quad \text{and} \quad \hat{x} = \sqrt{\frac{1}{\tau} \int_0^\tau (x)^2 d\tau} \quad (10)$$

Fourth order Runge-Kutta method is employed for numerical simulations, and the following parameters are adopted for all simulations: $\alpha = -0.5$ and $\beta = 0.1$, $\zeta = \zeta_{s1} = \zeta_{s2} = 0.02$, $\epsilon = 0.05$, $\rho = 20$ and, $\sigma = 0.5$ and $\chi = 0.1$ (De Paula et al., 2015). The dimensionless forcing frequency, v_{s1} and v_{s2} associated with dimensionless linear stoppers stiffness and γ_1 and γ_2 associated with dimensionless distances between the stoppers and vibrating mass are varied to evaluate the system efficiency.

3. RESULTS

The first analysis was carried out by selecting the Duffing parameters as $\alpha = -0.5$ and $\beta = 0.1$. These values were chosen to ensure a bistable Duffing response, represented by a double-well potential, which is the most relevant configuration for investigating the interaction between nonlinear restoring effects and impact-induced non-smooth dynamics. This choice is supported by previous studies on nonlinear piezomagnetoelastic energy harvesters, where the Duffing coefficients determine the potential energy profile and the corresponding dynamic regime (De Paula et al., 2015). Figure 3(a) shows restitution force in function of displacement and Figure 3(b) shows the potential energy.

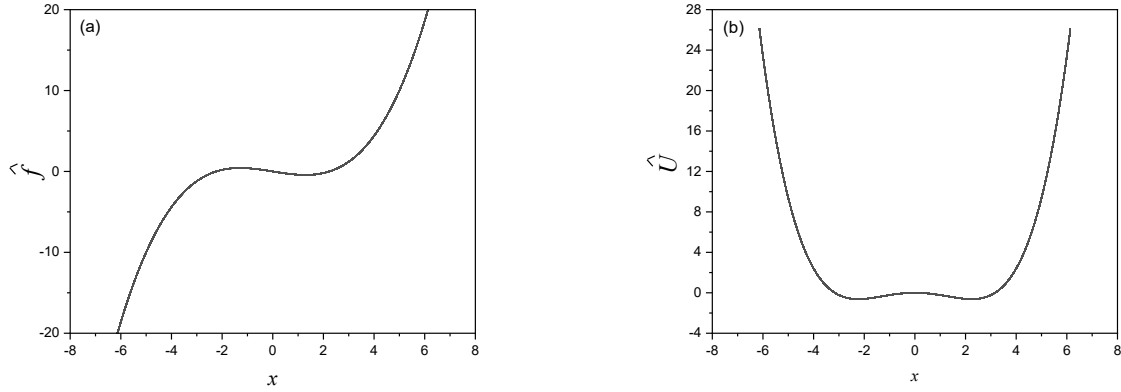


Figure 3. Duffing type system: (a) force and (b) potential energy for the values of α and β .

To further investigate the nonlinear dynamics of the system, the phase space trajectories are analyzed. Figure 4 presents the phase portrait of the Duffing-type oscillator for different values of the stopper stiffness while maintaining the same gap distance. The trajectories represent the relationship between displacement and velocity, allowing a better understanding of the dynamic behavior of the system. The dashed vertical lines indicate the position of the mechanical stoppers highlighting the limits where impact interactions occur. These impacts introduce non-smooth transitions in the system dynamics, which alter the trajectories in the phase plane and influence the overall response of the system.

It can be observed that, in the absence of mechanical stoppers, the system presents larger phase trajectories, which correspond to higher displacement amplitudes. When the mechanical stoppers are introduced, the trajectories become confined within region defined by the impact boundaries. This confinement limits the oscillation amplitude and modifies the structure of the phase space. The discrete points shown in Figure 4 represent the Poincaré sections, obtained by sampling the system state once per excitation period. These points provide a complementary view of the phase-space trajectories, helping to identify the structure and regularity of the system response.

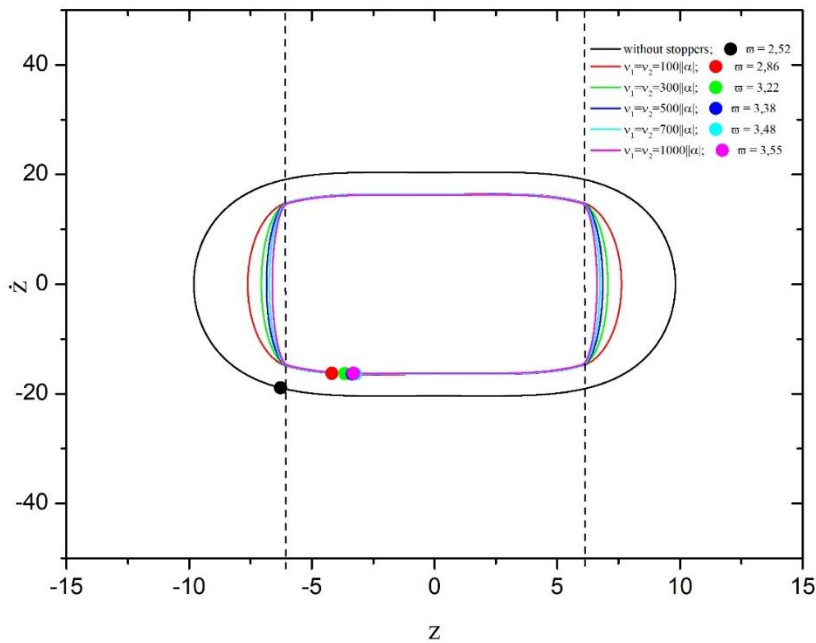


Figure 4. Phase space trajectories $(z, \dot{z})$ of the Duffing-type energy harvesting system for different stopper stiffness values ($\alpha = -0.5$, $\beta = 0.1$).

Figure 5(a) presents the results as function of the excitation frequency of the average dimensionless displacements without and with stoppers with different stiffness values while maintaining the same gap distance. It can be observed that the impacts lead to resonance peaks becoming broader and less sharp, this behavior becomes more pronounced as the support stiffness increases. Figure 5(b) shows the same results but now under the perspective of average dimensionless power output. The impacts excite some frequency components increasing the system bandwidth. Nevertheless, increasing the stiffness of stoppers also restricts the displacement and power amplitudes.

To compare the power gain introduced by the impacts in the Duffing system, Figure 5(b) highlights a region delimited by $\Delta\bar{\omega}_1$. Within this region, the system with impacts exhibits superior performance. Outside this interval, the Duffing system without impacts is consistently more efficient, but shows a considerable reduction in efficiency within the $\Delta\bar{\omega}_1$ range. Figure 6 shows the analysis considers average values of $\hat{P}_{out}$ between $\Delta\bar{\omega}_1$, called $\bar{P}_{out}$. Note that this approach allows one to evaluate the average energy harvesting capacity through a frequency range, all the cases with impacts increase the power in comparison with system having only the duffing characteristics.

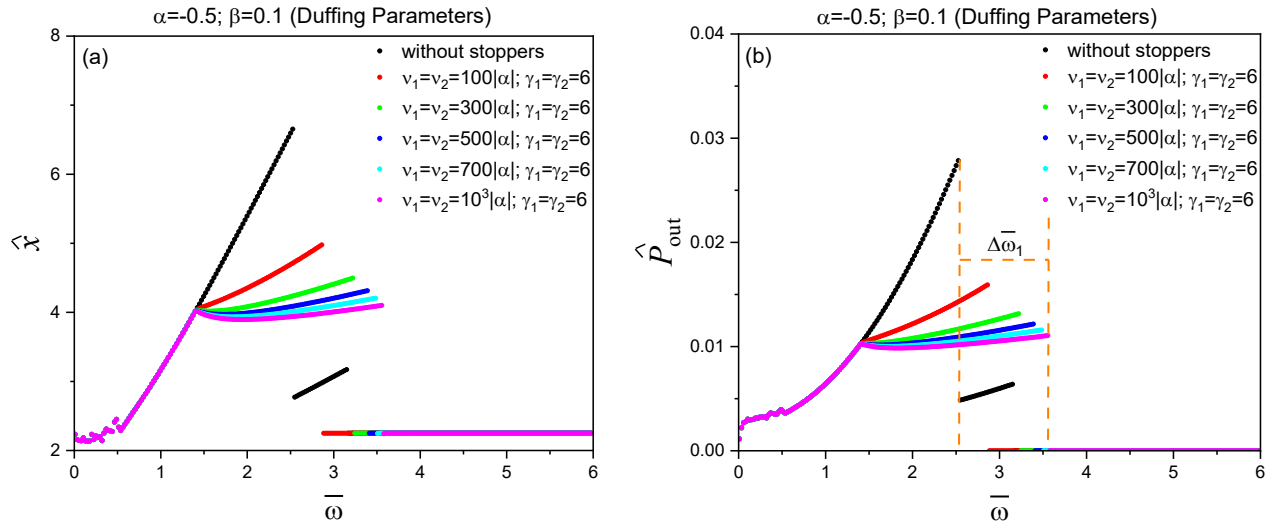


Figure 5. Duffing type system with stoppers with different stiffness values and same gap distances: (a) dimensionless displacement and (b) dimensionless power output

Figure 6 shows the analysis considering the dimensionless average values of $\hat{P}_{out}$ between $\Delta\bar{\omega}_1$, called $\bar{P}_{out}$. Note that this approach allows one to evaluate the average energy harvesting capacity through a frequency range, all the cases with impacts increase the power in comparison with system having only the duffing characteristics.

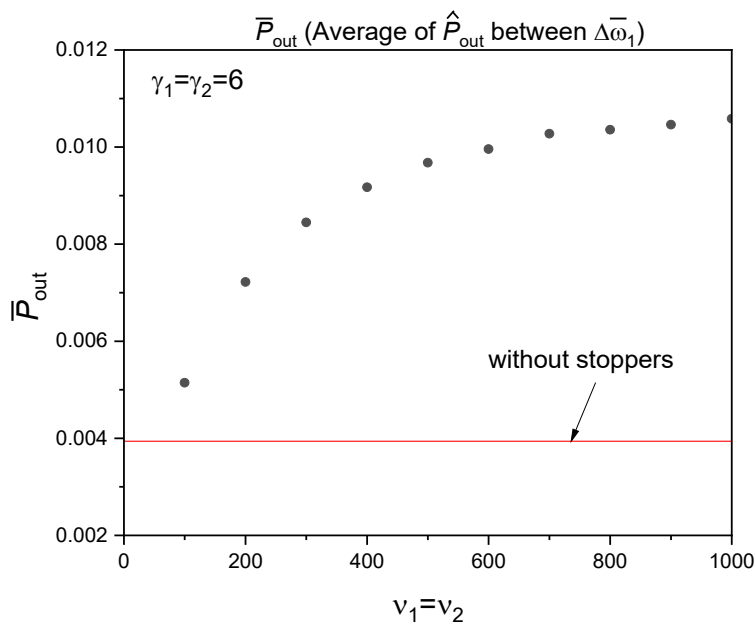


Figure 6. Dimensionless average harvested power output for $\Delta\bar{\omega}_1$ interval for different stiffness values.

Figure 7 presents the results as function of the excitation frequency of the average dimensionless displacements (a) and average power (b) without and with stoppers with different gap values while maintaining the same stiffness. These curves show that the same conclusions related to displacement and power previously observed.

To compare the power gain introduced by the impacts in the Duffing system, Figure 7(b) highlights a region delimited by $\Delta\bar{\omega}_2$ to compare the performance of both systems without and with impacts. Figure 8 shows that the cases with impacts present higher average values of $\hat{P}_{out}$ between $\Delta\bar{\omega}_2$, called $\bar{P}_{out}$, in comparison with system having only the duffing characteristics, excluding the condition with a gap of 8.

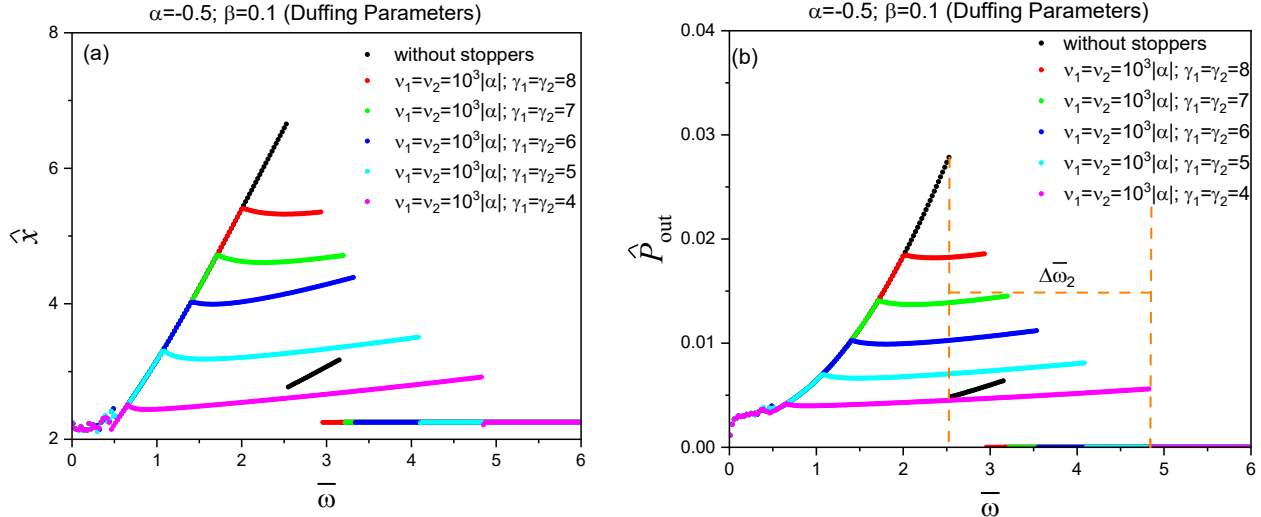


Figure 7. Duffing type system with stoppers with different gap distances and same stiffness values: (a) dimensionless displacement and (b) dimensionless power output

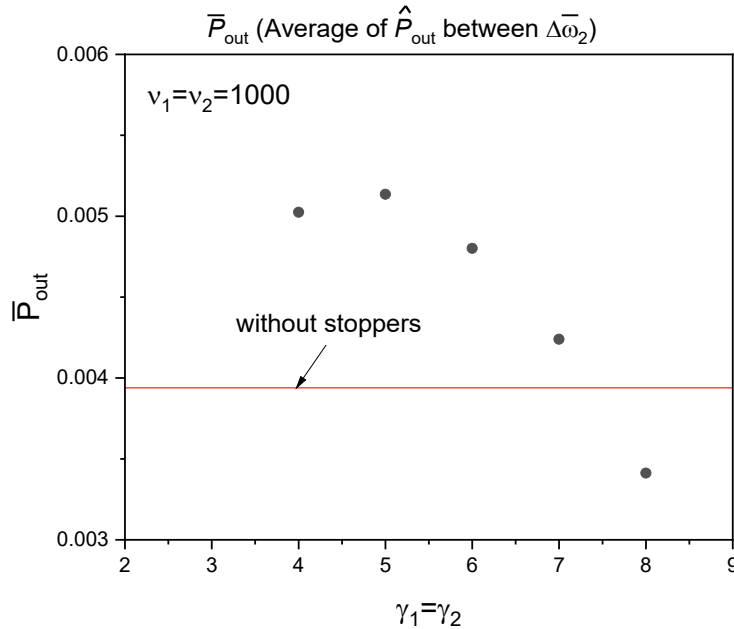


Figure 8. Dimensionless average harvested power output for $\Delta\bar{\omega}_2$ interval for different gap values.

4. CONCLUSIONS

This work investigates the dynamic behavior and energy harvesting performance of a piezoelectric system exhibiting Duffing-type and nonsmooth nonlinearities. A single-degree-of-freedom mechanical model is formulated by combining the linear piezoelectric constitutive relations including a Duffing term and two mechanical end-stops. The numerical results indicate that the introduction of Duffing-type nonlinearity with the traditional non-smooth systems can significantly broaden the effective frequency bandwidth of operation and increase the average harvested power over a

range of excitation conditions. Therefore, the introduction of impacts may be an interesting strategy when the system operates within this specific frequency interval.

5. ACKNOWLEDGEMENTS

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7. AUTHORSHIP RESPONSIBILITY

The authors are solely responsible for the content of this work.